



SILVER OAK UNIVERSITY
Silver Oak College of Computer Application
Integrated Master Of Science Information Technology
Course Name: Ethical and Legal Frameworks for Digital
Systems
Course Code: MDC5011C
Semester: 7th

Prerequisite:

1. Basic understanding of Information Technology and Computer Networks
2. Fundamentals of Cybersecurity
3. Awareness of Digital Technologies and Internet-based Services

Course Objectives:

1. To introduce ethical principles and professional responsibilities in digital environments.
2. To understand cyber laws, privacy regulations, and intellectual property rights governing digital systems.
3. To analyze ethical, legal, and regulatory challenges associated with cybersecurity and emerging technologies.
4. To apply governance, compliance, and legal frameworks in securing digital systems and information assets.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
4	60	0	0	0	0	40	20	120	4

Contents:

Unit	Topics	Teaching Hours	% Weightage
1	Introduction to Ethics and Digital Citizenship Introduction to Ethics and Morality, Importance of Ethics in Digital Society, Professional Ethics in Information Technology, Responsible Use of Computers and Internet, Digital Citizenship and Online Behavior, Digital Footprint and Online Reputation, Cyberbullying, Online Harassment, and Digital Responsibility, Ethical Issues in Social Media Usage, Ethical Decision-Making in Everyday Digital Activities	15	25

2	Fundamentals of Cyber Laws and Privacy Introduction to Cyber Law, Need for Cyber Laws in Digital Society, Overview of Information Technology Act, 2000, Common Cyber Crimes and Legal Implications, Digital Signatures and Electronic Records, Introduction to Privacy and Data Protection, Personal Data and Sensitive Information, Digital Personal Data Protection Concepts, Rights and Responsibilities of Digital Users	15	25
3	Intellectual Property Rights and Emerging Digital Technologies Introduction to Intellectual Property Rights (IPR), Copyrights, Patents, Trademarks, and Trade Secrets, Software Piracy and Digital Content Protection, Open Source Software and Licensing Basics, Introduction to Artificial Intelligence (AI), Ethical Use of AI Systems, AI Bias and Fairness Concepts, Ethical Issues in Social Media Algorithms, Privacy Concerns in AI, Cloud Computing, and IoT	15	25
4	Responsible Technology Use and Digital Governance Information Security and User Responsibilities, Organizational Policies and Code of Conduct, Cybersecurity Awareness and Best Practices, Ethical Handling of Information and Data, Responsible Use of Emerging Technologies, Digital Inclusion and Accessibility, Sustainability and Social Impact of Technology, Introduction to Cybersecurity Governance, Future Trends in Digital Ethics and Technology Regulation	15	25
Total		60 Hours	100%

Course Outcomes:

CO No.	CO Statement	Blooms Taxonomy Level	Unit
CO-1	Explain fundamental ethical principles, professional responsibilities, and digital citizenship practices in the use of information systems.	U	1,4
CO-2	Describe basic cyber laws, privacy regulations, and intellectual property rights applicable to digital systems.	U	2,3
CO-3	Analyze ethical and legal issues associated with cybersecurity incidents, social media, digital platforms, and emerging technologies.	AN	2,3
CO-4	Apply ethical decision-making and responsible technology practices to address challenges in digital environments.	A	1,2,4

Teaching & Learning Pedagogy approaches:

Interactive Lectures and Classroom Discussions
 Case Study Analysis on Cyber Ethics and Digital Laws
 Problem-Based Learning (PBL)

Group Discussions and Debates on Ethical Issues
 Flipped Learning Activities
 Collaborative Learning and Team-Based Exercises
 Scenario-Based Learning and Role Playing
 Research and Presentation Activities
 Multimedia-Based Learning (Videos, Podcasts, Online Resources)
 Self-Learning and Reflective Learning Activities

List of Tutorials: NA

Sr. No.	Tutorial Content	Hours	Co No.

List of Experiments: NA

Sr. No.	Practical Name	Hours	Co No.

Major Equipment and Simulation Tools:

1. Computer System with Internet Connectivity
2. Learning Management System (LMS)
3. Presentation and Collaboration Tools
4. Cyber Law and Regulatory Portals
5. Digital Privacy and Compliance Resources
6. AI Ethics and Responsible AI Demonstration Tools
7. Case Study Repositories on Cyber Ethics and Digital Governance

8. Open-Source Learning Platforms (SWAYAM, NPTEL, Coursera, Cisco Networking Academy)

Books Recommended(APA Format):

1. Baase, S., & Henry, T. M. (2018). A gift of fire: Social, legal, and ethical issues for computing technology (5th ed.). Pearson Education.
2. Johnson, D. G. (2021). Computer ethics (5th ed.). Pearson Education.
3. Kizza, J. M. (2023). Ethical and social issues in the information age (7th ed.). Springer.
4. Quinn, M. J. (2020). Ethics for the information age (8th ed.). Pearson Education.
5. Reynolds, G. W. (2023). Ethics in information technology (7th ed.). Cengage Learning.
6. Sharma, V. (2018). Information technology law and practice: Cyber laws and laws relating to e-commerce. Universal Law Publishing.
7. Spinello, R. A. (2020). Cyberethics: Morality and law in cyberspace (7th ed.). Jones & Bartlett Learning.
8. Whitman, M. E., & Mattord, H. J. (2022). Principles of information security (7th ed.). Cengage Learning.
9. Greenleaf, G. (2024). Asian data privacy laws: Trade and human rights perspectives (3rd ed.). Oxford University Press.
10. Solove, D. J., & Schwartz, P. M. (2021). Information privacy law (7th ed.). Aspen Publishing.

Term Work Module:

40 HOURS

Objective: To develop awareness and understanding of ethical, legal, privacy, and governance issues in digital systems through case studies, discussions, presentations, and practical analysis of real-world scenarios.

Term Work	Statement	Hours	CO No.
Term Work 1	Case Study Analysis on Cyber Ethics and Digital Citizenship	08 Hours	CO1
Term Work 2	Analysis of Cybercrime Cases and Relevant Legal Provisions	08 Hours	CO2
Term Work 3	Seminar/Presentation on Emerging Ethical Issues in AI, Social Media, or Privacy	08 Hours	CO3
Term Work 4	Group Discussion/Debate on Contemporary Digital Ethics Topics	06 Hours	CO1,CO3, CO4
Term Work 5	Assignment on Intellectual Property Rights and Data Privacy	05 Hours	CO2, CO3
Term Work 6	Mini Project/Report on Responsible Technology Use and Digital Governance	05 Hours	CO1, CO4
Total Hours		40	

Self Learning Module:**20 HOURS**

Objective: To encourage independent learning and critical thinking regarding ethical, legal, and societal implications of digital technologies and emerging trends.

Recommended Online Resources:

Sr. No.	Unit Name/Topic Name	Resource Link
1	Cyber Ethics and Digital Citizenship	1. College Recorded Videos 2. NPTEL Courses 3. Microsoft Digital Citizenship and Online Safety Resources: https://www.microsoft.com/en-us/digital-skills 4. Cisco Networking Academy – Introduction to Cybersecurity: https://www.netacad.com/courses/cybersecurity/introduction-cybersecurity 5. Coursera – Ethics, Technology and Society Courses: https://www.coursera.org
2	Cyber Laws and Information Technology Act	1. College Recorded Videos 2. NPTEL Courses on Cyber Law and Cyber Security 3. SWAYAM Portal: https://swayam.gov.in 4. Ministry of Electronics and Information Technology (MeitY): https://www.meity.gov.in
3	Privacy and Data Protection	1. College Recorded Videos 2. NPTEL Courses 3. Digital Personal Data Protection Resources: https://www.digitalindia.gov.in 4. European Commission GDPR Resources: https://commission.europa.eu/law/law-topic/data-protection_en 5. Coursera Data Privacy Courses: https://www.coursera.org
4	Intellectual Property Rights (IPR)	1. College Recorded Videos 2. NPTEL Courses 3. WIPO Academy: https://www.wipo.int/academy/en/ 4. NPTEL Intellectual Property Rights Courses: https://nptel.ac.in

CO-PO-PSO Matrix:

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1	2	1	2	1	-	1	2	1	1
CO-2	2	2	1	-	1	1	2	1	1
CO-3	2	3	2	1	1	2	2	2	2
CO-4	2	3	2	2	1	2	3	2	2

CO-PO-PSO MAPPING JUSTIFICATION FOR ETHICAL AND LEGAL FRAMEWORKS FOR DIGITAL SYSTEMS

CO No.	CO Statement	PO/PSO No.	Justification
CO1	Explain fundamental ethical principles, professional responsibilities, and digital citizenship practices in the use of information systems.	PO1	Understanding ethical principles requires foundational knowledge of digital systems and computing concepts.
		PO2	Enables analysis of ethical issues and responsible behavior in digital environments.
		PO3	Promotes ethical awareness and professional responsibility in technology usage.
		PO4	Encourages evaluation of ethical practices and their impact on society.
		PO6	Develops effective communication of ethical responsibilities and professional conduct.
		PO7	Encourages lifelong learning, digital citizenship, and ethical self-reflection.
		PSO1	Supports ethical considerations during the design and development of computing solutions.

		PSO2	Enhances awareness of responsible technology use in emerging digital environments.
CO2	Describe basic cyber laws, privacy regulations, and intellectual property rights applicable to digital systems.	PO1	Requires understanding of legal frameworks governing digital technologies.
		PO2	Enables analysis of legal, privacy, and compliance requirements in information systems.
		PO3	Supports ethical and lawful application of technology.
		PO5	Develops awareness of governance, compliance, and organizational responsibilities.
		PO6	Enhances communication and interpretation of legal and regulatory requirements.
		PO7	Encourages continuous learning of evolving cyber laws and regulations.
		PSO1	Supports implementation of legal and regulatory requirements in IT systems.
		PSO2	Promotes awareness of legal considerations in emerging technologies and digital platforms.
CO3	Analyze ethical and legal issues associated with cybersecurity incidents, social media, digital platforms, and emerging technologies.	PO1	Applies foundational knowledge to understand ethical and legal issues in digital systems.
		PO2	Develops analytical skills for evaluating ethical and legal challenges.
		PO3	Encourages ethical reasoning and responsible decision-making in technology contexts.
		PO4	Supports investigation and evaluation of ethical and legal implications of digital incidents.

		PO5	Utilizes digital tools and platforms while considering ethical and legal constraints.
		PO6	Enhances communication and discussion of ethical and legal analyses.
		PO7	Promotes continuous learning regarding evolving ethical and legal issues.
		PSO1	Supports application of ethical and legal principles in computing solutions.
		PSO2	Develops understanding of ethical implications of emerging technologies and digital platforms.
CO4	Apply ethical decision-making and responsible technology practices to address challenges in digital environments.	PO1	Applies foundational knowledge to ethical decision-making in digital systems.
		PO2	Supports ethical problem-solving and responsible decision-making.
		PO3	Encourages development of ethical and socially responsible technology practices.
		PO4	Promotes collaborative evaluation and resolution of ethical challenges.
		PO5	Facilitates appropriate use of digital tools and technologies in compliance with ethical standards.
		PO6	Enhances communication and advocacy of responsible technology practices.
		PO7	Promotes reflective, independent, and lifelong professional practice.
		PSO1	Supports implementation of ethical principles in computing applications and systems.
		PSO2	Facilitates responsible use and management of emerging technologies.

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for Ethical and Legal Frameworks for Digital Systems (MDC5011C)		
SDG No. & Title	Relevance to Course	Justification
SDG 4: Quality Education	High	Promotes digital literacy, ethical awareness, digital citizenship, and responsible use of information systems. It equips students with knowledge of ethical, legal, and societal aspects of digital technologies.
SDG 9: Industry, Innovation and Infrastructure	High	Supports responsible innovation and sustainable digital infrastructure by addressing ethical and legal challenges associated with emerging technologies such as AI, cloud computing, IoT, and digital platforms.
SDG 16: Peace, Justice and Strong Institutions	High	Encourages ethical use of technology, privacy protection, cyber laws, intellectual property rights, digital governance, and regulatory compliance, contributing to secure and trustworthy digital ecosystems.
SDG 17: Partnerships for the Goals	Moderate	Promotes collaboration among individuals, organizations, policymakers, and technology providers to establish responsible and inclusive digital environments governed by ethical and legal frameworks.

Explanation

Ethical and Legal Frameworks for Digital Systems is a foundational course that introduces students to the ethical, legal, and societal aspects of modern digital technologies. The course focuses on ethical principles, digital citizenship, cyber laws, privacy protection, intellectual property rights, and responsible use of information systems. Students will explore ethical and legal challenges associated with cybersecurity, social media, artificial intelligence, and other emerging technologies through real-world examples and case studies. The course helps learners develop awareness of professional responsibilities, informed decision-making skills, and a strong ethical foundation required for careers in information security and related technology domains.